$$\lambda x.\ x(\lambda y.\ x(\lambda z.\ y))$$

$$\dfrac{}{x:\alpha_1,\ y:\alpha_4,\ z:\alpha_7 \vdash y:\alpha_8}\ \text{VAR}\ \boxed{6}$$

$$\dfrac{\dfrac{}{x:\alpha_1,\ y:\alpha_4 \vdash x:\alpha_6\to\alpha_5}\ \text{VAR}\ \boxed{4} \qquad \dfrac{x:\alpha_1,\ y:\alpha_4,\ z:\alpha_7 \vdash y:\alpha_8}{x:\alpha_1,\ y:\alpha_4 \vdash \lambda z.y:\alpha_6}\ \text{ABS}\ 5}{x:\alpha_1,\ y:\alpha_4 \vdash x(\lambda z.y):\alpha_5}\ \text{APP}$$

$$\dfrac{\dfrac{}{x:\alpha_1 \vdash x:\alpha_3\to\alpha_2}\ \text{VAR}\ \boxed{2} \qquad \dfrac{x:\alpha_1,\ y:\alpha_4 \vdash x(\lambda z.y):\alpha_5}{x:\alpha_1 \vdash \lambda y.x(\lambda z.y):\alpha_3}\ \text{ABS}\ \boxed{3}}{x:\alpha_1 \vdash x(\lambda y.x(\lambda z.y)):\alpha_2}\ \text{APP}$$

$$\dfrac{x:\alpha_1 \vdash x(\lambda y.x(\lambda z.y)):\alpha_2}{\vdash \lambda x.x(\lambda y.x(\lambda z.y)):\alpha_0}\ \text{ABS}\ \boxed{1}$$

$\boxed{1}$ $\alpha_0 = \alpha_1 \to \alpha_2$

$\boxed{2}$ $\alpha_1 = \alpha_3 \to \alpha_2$

$\boxed{3}$ $\alpha_3 = \alpha_4 \to \alpha_5$

$\boxed{4}$ $\alpha_1 = \alpha_6 \to \alpha_5$

$\boxed{5}$ $\alpha_6 = \alpha_7 \to \alpha_8$

$\boxed{6}$ $\alpha_4 = \alpha_8$

$\boxed{1}$ $\alpha_0 = \alpha_1 \to \alpha_2$

$\boxed{2}$ $\alpha_1 = \alpha_3 \to \alpha_2$

$\boxed{3}$ $\alpha_3 = \alpha_4 \to \alpha_5$

$\boxed{4}$ $\alpha_1 = \alpha_6 \to \alpha_5$

$\boxed{5}$ $\alpha_6 = \alpha_7 \to \alpha_8$

$\to$ $\boxed{6}$ $\alpha_4 = \alpha_8$

$\alpha_4 \longmapsto \alpha_8$

$\alpha_0 = \alpha_1 \to \alpha_2$

$\alpha_1 = \alpha_3 \to \alpha_2$

$\alpha_3 = \alpha_8 \to \alpha_5$

$\alpha_1 = \alpha_6 \to \alpha_5$

$\to$ $\alpha_6 = \alpha_7 \to \alpha_8$

$$\alpha_4 \longmapsto \alpha_8$$

$$\alpha_6 \longmapsto \alpha_7 \to \alpha_8$$

$$\alpha_0 = \alpha_1 \to \alpha_2$$

$$\alpha_1 = \alpha_3 \to \alpha_2$$

$$\alpha_3 = \alpha_8 \to \alpha_5$$

$$\to \alpha_1 = (\alpha_7 \to \alpha_8) \to \alpha_5$$

$$\alpha_4 \longmapsto \alpha_8$$

$$\alpha_6 \longmapsto \alpha_7 \to \alpha_8$$

$$\alpha_1 \longmapsto (\alpha_7 \to \alpha_8) \to \alpha_5$$

$$\alpha_0 = ((\alpha_7 \to \alpha_8) \to \alpha_5) \to \alpha_2$$

$$(\alpha_7 \to \alpha_8) \to \alpha_5 = \alpha_3 \to \alpha_2$$

$$\alpha_3 = \alpha_8 \to \alpha_5$$

$$\alpha_4 \longmapsto \alpha_8$$

$$\alpha_6 \longmapsto \alpha_7 \to \alpha_8$$

$$\alpha_1 \longmapsto (\alpha_7 \to \alpha_8) \to \alpha_5$$

$$\alpha_0 = ((\alpha_7 \to \alpha_8) \to \alpha_5) \to \alpha_2$$

$$\to (\alpha_7 \to \alpha_8) \to \alpha_5 = \alpha_3 \to \alpha_2$$

$$\alpha_3 = \alpha_8 \to \alpha_5$$

$$\alpha_4 \longmapsto \alpha_8$$

$$\alpha_6 \longmapsto \alpha_7 \to \alpha_8$$

$$\alpha_1 \longmapsto (\alpha_7 \to \alpha_8) \to \alpha_5$$

$$\alpha_0 = ((\alpha_7 \to \alpha_8) \to \alpha_5) \to \alpha_2$$

$$\alpha_7 \to \alpha_8 = \alpha_3$$

$$\alpha_5 = \alpha_2$$

$$\alpha_3 = \alpha_8 \to \alpha_5$$

$$\text{XOR } b_1\ b_2 =$$

$$\text{if } b_1 \text{ then not } b_2$$

$$\text{else } b_2$$

$$\text{if } b \text{ then } M \text{ else } N$$

$$\equiv \quad b\ M\ N$$

$$T = \lambda x \lambda y . x$$

$$F = \lambda x \lambda y . y$$

$$\text{XOR} = \lambda x \lambda y . x\ (\text{not } y)\ y$$

$$\text{not } b = \text{if } b \text{ then } F \text{ else } T$$

$$\text{not} = \lambda b . b\ F\ T$$

$$\neg\, T \longrightarrow_v^{*} F$$

$$(\lambda x.\ x\ F\ T)\,(\lambda x.\lambda y.x)$$

$$\longrightarrow_v \underbrace{\big((\lambda x.\lambda y.x)\ F\big)\ T}_{(\text{by } \beta)}$$

one small step

$$= (\bullet\ T)\,[(\lambda x.\lambda y.x)\ F]$$

$$\longrightarrow_v (\bullet\ T)\,[\lambda y.F]$$

$$= (\lambda y.F)\ T \quad \longleftarrow_v$$

$$\longrightarrow_v F \qquad \checkmark$$

$$(\text{xor } T)\ F$$

$$= (\bullet\ F)[\text{xor } T]$$

$$\to_v (\bullet\ F)[\lambda y.\, T\ (\text{not } y)\ y]$$

$$= (\lambda \underline{y}.\, T\ (\text{not } y)\ y)\ F$$

$$\cancel{= \bullet[(\lambda y.\, T\ (\text{not } y)\ y)\ F]}$$

$$\to_v T\ (\text{not } F)\ F$$

$$= (\bullet\ F)[T\ (\text{not } F)] \quad (\text{APP}_1)$$

$$= ((T\ \bullet)\ F)[\underbrace{\text{not}}_{(\lambda x.\, x F T)}\ F] \quad (\text{APP}_2)$$

$$\to_v ((T\ \bullet)\ F)[F\ F\ T]$$

$$= (T\ (F\ F\ T))\ F$$

$$= ((T\ (\bullet\ T))\ F)[F\ F]$$